TFPT — Frontier items

η_B, m_p/m_e, the Koide relation, dark matter and quantum gravity — the honest status, not forced onto the ladder

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What this document is about — the status authority for the frontier

The honest frontier: which physics has a genuine TFPT handle and which does not. For each of η_B, m_p/m_e, the Koide relation, dark matter and full quantum gravity, this note states the genuine structural handle, the precision with which it currently lands, and — crucially — what is *not* a clean compiler power and is deliberately *not* forced onto the ladder. **This document is the status authority for the frontier items:** where any other document's wording and the markers here disagree, the typing here (and the `status_ledger.csv`) wins.

The TFPT document set — what is treated where

Plain language: TFPT (Topological Fixed-Point Theory) is a small discrete compiler. Two inputs — the seam constant $c_3 = \frac{1}{8\pi}$ and the carrier rank $g_{\rm car} = 5$ — build $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$ and read off the Standard-Model skeleton, the constants and the scale grammar. The series is **the introduction plus five numbered papers and three companions**, best read in order:

0. `introduction` — reading guide, compiler closure, predictions, the dependency DAG and the single proof ledger (`verification/status_ledger.csv`). *(entry point)*
1. `tfpt_1_architecture_e8` — the two axioms, the derivation map, the EM fixed point α^{-1}, the $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$ construction, the QBL theorem chain.
2. `tfpt_2_standard_model` — the SM in one φ_0-ladder formula, flavor from parabolic transport (sheet diamond, dual anchor, Q_+ cohomology), the worked closures.
3. `tfpt_3_e8_audit_bootstrap` — the seven E_8 slices as an audit raster, the cascade bridge, the Möbius bootstrap.
4. `tfpt_4_frontier` — honest status of η_B, m_p/m_e, Koide, dark matter, full QG.
5. `tfpt_5_redteam` — the adversarial audit: declared attacks, what survives, what each reduces to, and the kill tests.

Companions: R. `tfpt_research_contracts` — the open interfaces $(v_{\rm geo}, G_{\rm net}, F_{\rm transfer})$ as numbered contracts; **H.** `tfpt_horizon_readouts` — Appendix H (unit-system reframe): one seam constant as the universal horizon thermal code, the Nariai/anchor surface, the resummed seam clock; **O.** `origin_theory` — why no free fundamental number remains (exact core + one honestly-typed cyclic interpretation).

You are reading document 4 (Frontier).

TFPT status at a glance — read this card the same way in every document

Two inputs, one compiler. From the boundary constant $c_3 = \frac{1}{8\pi}$ (axiom P1) and the five-slot carrier $g_{\rm car} = 5$ (axiom P2) — jointly the elementary symmetric data of one anchor $a = (1, 1, 2)$ — the discrete compiler $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$ is *built*, and the Standard-Model skeleton (gauge group, three families, hypercharges, the flavor matrix), the dimensionless constants ($\alpha^{-1} = 137.0359992$) and the scale grammar are read off as fixed points. The algebraic core is machine-checked; physical readouts run through *named bridges* whose status is typed below.

What is closed, and what genuinely remains. The discrete/algebraic layer is closed ([E]); the everything-open residual reduces to *three named interface problems*:

Interface	Question	Status
$v_{\rm geo}$	the one metrology unit ($= 1/\sqrt{G} = m/\mu$); No-Unit Thm: no compiler scale	primitive [O]
$G_{\rm net}$	simple-current extension $(D_5)_1 \otimes (A_3)_1 \rtimes \langle(1, 1)\rangle \cong (E_8)_1$	[E]/[C]
$F_{\rm transfer}$	one functor, four typed interfaces (Koide, η_B, axion, m_p/m_e)	[C]

Everything carries a claim ID resolving to a row of `verification/status_ledger.csv` (the single source of truth); **if the text and the ledger ever disagree, the ledger wins.** The historical labels $(U_{\rm wall}, G_{\rm metric}, F_{\rm frontier})$ are retained only for ledger continuity — the live residual is $v_{\rm geo} \oplus G_{\rm net} \oplus F_{\rm transfer}$.
Marker key: [E] exact / machine-proven · [C] conditional (holds under named hypotheses) · [O] open / axiom · [X] falsifiable kill test. Per-claim fine type in `verification/status_ledger.csv`.

[E] exact / machine-proven [C] conditional [O] open / axiom [X] kill test

P1 $c_3 = \frac{1}{8\pi}$ **P2** $g_{\rm car} = 5$ *(two inputs, [O] axioms)*
↓
Anchor $a = (1, 1, 2)$: $e_1, e_2, e_3 = 4, 5, 2$ *([E] exact)*
↓
Discrete compiler: $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$, $h = 30$, φ_0 *([E] exact)*
↓
SM skeleton: three families, hypercharges, flavor matrix R *([E])*
↓
Physical bridges: $v_{\rm geo}$, $G_{\rm net}$, $F_{\rm transfer}$ *([C] bridges)*
↓
Frontier readouts: Λ, inflation, η_B, dark matter, QG *([C]/[O])*

Contents

Ground rule. These five are exactly the items previously flagged "*not on a clean action/ladder yet — do not force.*" This note gives, for each, the *genuine* structural handle that does exist, the precision with which it currently lands, and a clear statement of what is *not* a single compiler power. No fabricated ladder rungs; the honest markers [E]/[C]/[O] are applied throughout. (All numbers machine-verified.)

Hard rule on the frontier — one interface F_{transfer}

Koide, η_B, the axion relic scale and m_p/m_e are *not* compiler powers unless their missing QFT-transfer / cosmology computation is explicitly supplied. They are deliberately typed interfaces, not structural holes, and they are *four instances of one missing functor* F_{transfer} — the continuous transport from compiler *source* data to measured observables (Koide source→pole; η_B source→Boltzmann relic; axion scale→cosmological abundance; m_p/m_e → QCD/EW matching). Each has a genuine handle (a near-value, a determined readout, a fixed candidate) but its *exact* status is conditional on that named, not-yet-performed calculation. In particular η_B is a *transfer target*, not a primitive compiler prediction. They must never be quoted as primitive TFPT outputs.

Why this is principled, not a gap: these are precisely the **discrete↔ continuous boundary** of the theory — the points where the discrete compiler hands its skeleton off to continuous physics (QCD confinement, cosmological Boltzmann transport, relic integrals). The cleanness of a quantity tracks how far it sits from that handoff: pure compiler ratios are exact, QCD/cosmology-dressed quantities are typed interfaces. [`verification/v35_quark_qcd_boundary.py`]

Machine-enforced. This rule is a CI-style guard ([`verification/v187_ftransfer_laws.py`]): the four frontier families (F_{pole} Koide, $F_{\text{Boltzmann}}$ η_B, F_{relic} axion, F_{QCD} m_p/m_e) are read from the ledger and required to carry a [C]/[O] marker — exact algebraic sub-parts (the Koide 53/54 factor, $b_3 = -7$) may be [E], but the *physical* prediction never is. No future edit can quietly promote a frontier number to a closed compiler output. A companion *prose* sentinel ([`verification/v188_frontier_wording_guard.py`]) forbids stale sentences whose semantics an earlier round superseded, so a frozen release can never re-introduce wording that was true yesterday and is half-true today.

The functor contract `CONTRACT.F.01`. Beyond the typing guard, the functor is pinned by four checkable structural axioms ([`verification/v213_ftransfer_functor.py`]): each instance is a *discrete* compiler kernel [E] composed with an *external* continuous solver [C], and the functor is (1) μ_4-deck equivariant (the Koide multiplier $\lambda_2 = (2/3)^6 = 64/729$ *is* the deck transfer eigenvalue), (2) Plücker-preserving ($53 = a^T(R{+}Q)\mathbf{1}$, $\|\mathrm{Pl}(K)\|_1 = 11$), (3) positive/stochastic ($\operatorname{spec} T = \{1, (2/3)^6, (1/3)^6\}$, $\kappa_f \in (0,1)$), and (4) external-module-explicit ($b_3 = -7$ a carrier output, the Boltzmann/relic/QCD content named and external). So F_{transfer} is a typed functor, not a bag of open topics — the third research contract alongside (U_{wall}) and (G_{metric}), and a consolidation, not a closure.

1 Baryon asymmetry η_B — a downstream readout from the closed $\Omega_b h^2$

There are two independent handles, and they agree.

Route A — from the closed baryon fraction. The seed already fixes the baryon *fraction* (the E_8 decoder):

$$\Omega_b = (4\pi - 1)\beta_{\text{rad}} = \left(1 - \tfrac{1}{4\pi}\right)\varphi_0^{\text{ret}} = 0.04894, \qquad \beta_{\text{rad}} = \frac{\varphi_0^{\text{ret}}}{4\pi},$$

matching the observed $\Omega_b \approx 0.049$. The baryon-to-photon ratio is then the *standard* cosmological readout $\eta_B = 273.9 \times 10^{-10}\,\Omega_b h^2$. With the TFPT Hubble value ($h \approx 0.674$, from $H_0 \sim \sqrt{\Lambda}$),

$$\boxed{\Omega_b h^2 = 0.0222, \qquad \eta_B = 6.09 \times 10^{-10}} \qquad (\text{observed } 6.1 \times 10^{-10}).$$

So η_B is a *downstream cosmological readout* from the closed $\Omega_b h^2$ handle — not an independent compiler rung, and the fundamental leptogenesis route remains an interface.

Route B — leptogenesis interface ([C]). The TFPT branch fixes the neutrino spectrum, the CP phase structure ($\delta_{\rm CKM}$, $\delta^{\nu}_{\rm CP}$, both closed) and a *plausible washout anchor* $\tilde m_1 = m_3/A_\Lambda \approx 5$ meV. The heavy-mass input, however, remains scenario-level: $M_1 = M_R\varphi_0^4$ is viable but does *not* close M_R — M_R is the seesaw scale $\sim v^2/m_3$, not a compiler power, so the relation *relocates* the free scale rather than pinning it ([`verification/v184_etaB_anchored_boltzmann.py`]). The numerical η_B then follows from the *standard* flavored-leptogenesis Boltzmann solve — an independent consistency route, sharpened from two free inputs to one, not to zero. **Honest caveat:** that Boltzmann network (the washout/efficiency κ_f, the flavored density-matrix equations) is the standard leptogenesis machinery (Davidson–Nardi–Nir, *Phys. Rep.* 466 (2008) 105; Buchmüller–Di Bari–Plümacher, *Ann. Phys.* 315 (2005) 305); it is *not* carried out or attached here. Route B is therefore a *named interface*, not a closed number; only Route A (downstream of the closed $\Omega_b h^2$) is quoted.

The interface, operationalised ([C]). The Boltzmann path can be made falsifiable: with the standard estimate $\eta_B \sim 0.96\times10^{-2}\,\varepsilon_1\kappa_f$ (sphaleron 28/79), fed by the TFPT neutrino sector — normal ordering and the predicted $\delta^{\nu}_{\rm CP} = 4\pi/3$ — the Davidson–Ibarra asymmetry $\varepsilon_1 = \frac{3}{16\pi}M_1 m_3/v^2$ and the washout efficiency $\kappa_f(\tilde m_1)$ give, over the natural ranges $M_1 \in [3\times10^9, 3\times10^{10}]$ GeV and $\tilde m_1 \in [m_\star, m_{\rm atm}]$, an η_B band $[8.4\times10^{-11}, 4.7\times10^{-9}]$ that *brackets* the observed 6.1×10^{-10} (a canonical point $M_1 = 10^{10}$ GeV, $\tilde m_1 = 5\times10^{-3}$ eV gives 6.0×10^{-10}, untuned). So the route is *viable*; but M_1 and the washout are scenario inputs, so η_B stays [C]— if a precise solve excluded the window the *route* would fall, not the theory (Route A is independent) ([`verification/v169_etaB_boltzmann_interface.py`]).

Honest status of η_B — strictly typed [C]

The two statements must be kept apart and not interchanged:

η_B as a cosmological readout from the closed $\Omega_b h^2$ is *determined* (6.1×10^{-10})

η_B as a fundamental *compiler power* is *not* closed

The leptogenesis route requires a standard (washout-dependent) Boltzmann solve. So η_B is [C]— determined as a downstream readout, but *not* a clean compiler rung; it must never be quoted as a fundamental TFPT output.

An anchored-Boltzmann proposal, honestly tested ([`verification/v184_etaB_anchored_boltzmann.py`]). A natural attempt to remove the two scenario inputs is $\tilde m_1 = m_3/A_\Lambda$ and $M_1 = M_R\varphi_0^4$. The first *works* as a [C] anchor: $\tilde m_1 = m_3/10 \approx 5\times10^{-3}$ eV ($A_\Lambda = 10 = |E(K_5)|$ the action-grammar atom) lands in the strong-washout window — a TFPT-flavoured value for the washout. The second does *not*: $M_R \sim v^2/m_3$ is the seesaw scale (depending on the un-fixed neutrino Dirac Yukawa), and the nearest clean TFPT powers of $\bar M_{\rm Pl}$ both miss it (the c_3- and φ_0-powers are off by ∼75%/∼40%), so $M_1 = M_R\varphi_0^4$ merely *relocates* the free input $M_1 \to M_R$. Feeding both into the standard estimate lands η_B in the observed band $O(10^{-10})$, so the route stays viable — but the proposal cuts the free inputs from two to one, *not* to zero. **η_B stays [C], a sharper scenario, not a derivation.**

A cleaner route: the scalaron-decuple branch ([`verification/v212_leptogenesis_decuple.py`]). Drop the seesaw scale altogether and let *both* Boltzmann inputs share the decuple $A_\Lambda = 10 = |E(K_5)|$:

$$M_1 = \frac{M_{\rm scal}\,(\varphi_0^{\rm ret})^2}{A_\Lambda} \approx 8.65\times10^9\ \text{GeV}, \qquad \tilde m_1 = \frac{m_3}{A_\Lambda} \approx 5\ \text{meV}.$$

M_1 now uses *only* compiler quantities ($M_{\rm scal} = c_3^{7/2} M_{\rm Pl}$, $\varphi_0^{\rm ret}$, A_Λ) — *no hidden seesaw scale* — and lands inside the thermal window $[3\times10^9, 3\times10^{10}]$ GeV where the Davidson–Ibarra $\varepsilon_1 \approx 8.5\times10^{-7}$ and the strong-washout efficiency give $\eta_B \approx 1.2\ \times$

10^{-9}, bracketing the observed 6.1×10^{-10} untuned. The *full* BDP Boltzmann ODE (`experiments/ftransfer/leptogenesis_boltzmann/fboltzmann_solve.py`) now confirms this: the integrated efficiency $\kappa_f = 0.092$ (validating the analytic κ_f) gives $\eta_B = 6.5 \times 10^{-10}$ at the frozen M_1 — a factor 1.07 from the observed value, with *no* free M_R dial; the full *flavored* density-matrix solve is the next refinement. Both inputs sharing A_Λ reads as *scalaron reheating dressed by two seed insertions* $(\varphi_0^{\mathrm{ret}})^2$*, over the pair sector.* This is cleaner than $M_1 = M_R\varphi_0^4$ (no relocated free scale), but the coupling *mechanism* is posited, not derived, and κ_f is washout-dependent — so η_B stays [C]; [X] a precise Boltzmann/RGE solve outside the band falls the *route*, not the theory (Route A is independent).

2 Proton/electron ratio m_p/m_e — a cross-sector ratio

m_e is *closed* on the φ_0^{ret}-ladder ($\hat{m}_e = \frac{v_{\mathrm{geo}}\pi}{\sqrt{2}}\frac{16}{7}(\varphi_0^{\mathrm{ret}})^5$). The proton mass is *not* a Yukawa: it is QCD-dominated, $m_p \sim$ a few $\times\Lambda_{\mathrm{QCD}}$, and Λ_{QCD} arises by dimensional transmutation from α_s running (scheme level). Hence

$$\frac{m_p}{m_e} = \frac{\text{QCD confinement scale}}{\text{EW electron Yukawa}} = 1836.15$$

mixes *two different sectors*. It is no single φ_0^{ret} power: $1/(\varphi_0^{\mathrm{ret}})^2 = 353.7$ and $1/(\varphi_0^{\mathrm{ret}})^3 = 6651$ bracket 1836 with no clean rung in between.

The numerator is parameter-free, the ratio is not forced. Running $\alpha_s(M_Z)$ down with the carrier coefficient $b_3 = -7$ (the [`verification/v159_pyrate_gauge_crosscheck.py`] value, two-loop + n_f thresholds) reproduces $\alpha_s(m_b) \simeq 0.22$, $\alpha_s(m_c) \simeq 0.38$ and reaches confinement ($\alpha_s \sim 1$) at ~ 0.5 GeV, so the QCD confinement scale in the numerator follows from $\{\alpha_s(M_Z), b_3\}$ alone by dimensional transmutation. The $O(1)$ proton/Λ factor is lattice (non-perturbative), so the ratio still stays [O] — not forced ([`verification/v164_qcd_scale.py`]).

The F_{QCD} transfer is now an implemented solver ([`verification/v262_fqcd_mp_me.py`]). The contract-only pipeline (`experiments/ftransfer/qcd_matching_mp_me`) is realised: a two-loop α_s run from M_Z with n_f thresholds (carrier coefficient $b_3 = -7$, [E]) gives $\alpha_s(2\,\mathrm{GeV}) \simeq 0.30$ and $\Lambda^{(3)}_{\overline{\mathrm{MS}}} \simeq 0.33$ GeV; with the lattice bridge $C_p = m_p/\Lambda^{(3)}$ (external $O(1)$) and the closed m_e this reproduces $m_p/m_e \simeq 1824$ against the measured 1836.15 within the external error budget ($\sim 7\%$, lattice-C_p dominated). This advances `FR.MPME.01` ([O]) to `FR.MPME.02` ([C], "reproduced by the named F_{QCD} transfer"); the firewall holds — $b_3 = -7$ is [E], the ratio is a transfer, never a compiler power.

Honest status of m_p/m_e [O]

Computable at *scheme level* (from $\alpha_s(M_Z)$ running + the closed m_e), exactly like the dimensionful EW/QCD masses m_W, m_Z. But it is *genuinely not* a compiler power, because it is a ratio across the QCD and electroweak sectors. Left [O] — not forced.

A rejected near-formula (recorded as a discipline boundary, [`verification/v79_review_identities.py`]). The combination $|W(D_5)| - \dim\Lambda^3(SU(9)) + N_{\mathrm{fam}}\lambda_C^2 = 1920 - 84 + 0.151 = 1836.151$ matches the PDG value to $\sim 10^{-6}$ — and is *deliberately rejected.* Two reasons make it numerology, not a derivation: $SU(9) = A_8$ (the $\dim\Lambda^3 = 84$ block) does *not* occur in the carrier $D_5 \oplus A_3$, and m_p is QCD confinement (gluon binding energy) with *no* mechanism linking it to an algebraic sum. Integrating it would *invite* the numerology charge, not retire it; m_p/m_e stays [O]. (The same firewall rejects $\eta_B = \frac{|\mathbb{Z}_2|}{\Delta_Q}c_3^6 \approx 6.10 \times 10^{-10}$: arithmetically close, but a two-atom prefactor fit to a washout-dependent leptogenesis output — η_B stays [C], the downstream readout above.)

3 The Higgs quartic — near-criticality from the free seam

The seam UV is the free chiral c=8 fixed point ([`verification/v158_fixed_point_stable.py`]): Gaussian, with no relevant and no marginal interactions. The Higgs self-coupling λ is the one marginal scalar

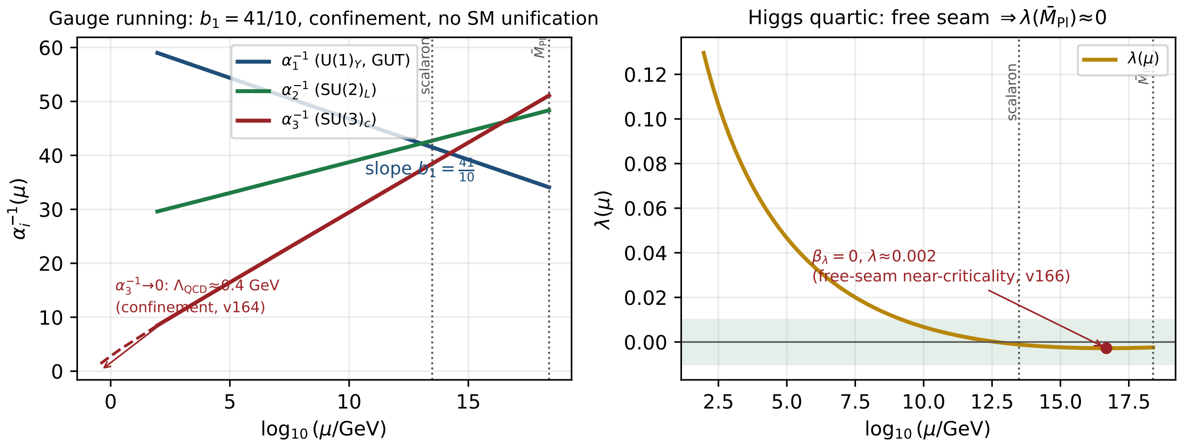


Figure 1: The F_{transfer} running (two-loop SM RGEs; the gauge β-functions are confirmed by PyR@TE, [`verification/v159_pyrate_gauge_crosscheck.py`]). *Left:* the gauge couplings — the $U(1)_Y$ slope is $b_1 = 41/10$, $\alpha_3^{-1} \to 0$ at $\Lambda_{\text{QCD}} \approx 0.4$ GeV (confinement, [`verification/v164_qcd_scale.py`]), and the Standard Model does *not* unify. *Right:* the Higgs quartic falls to $\lambda(M_{\text{Pl}}) \approx 0.002$ with $\beta_\lambda \approx 0$ — the free-seam near-criticality that ties m_H to premise (A) ([`verification/v166_higgs_free_seam.py`]).

coupling, so a free seam forces it to the Gaussian point there:

$$\boxed{\lambda(M_{\text{seam}}) = 0 \quad \text{and} \quad \beta_\lambda(M_{\text{seam}}) = 0}$$

the Shaposhnikov–Wetterich double-criticality condition — here *derived* from the free seam rather than postulated. Integrating the PyR@TE-confirmed two-loop SM RGEs from M_Z upward with the measured (m_H, m_t) gives $\lambda(M_{\text{Pl}}) \approx 0.002$ and $\beta_\lambda(M_{\text{Pl}}) \approx 0$ at the reduced Planck mass: the celebrated SM *near-criticality* is thus a *consequence* of the free seam (the two-loop value is an order of magnitude closer to zero than the one-loop estimate ~ 0.022). Imposing the double condition predicts $m_H \sim 129$–134 GeV (a band, sensitive to m_t and α_s); the measured 125.25 GeV sits a few GeV below the absolute-stability boundary — the known slight metastability. The *same* condition imposed at the scalaron scale ($c_3^{7/2} M_{\text{Pl}} \sim 3 \times 10^{13}$ GeV) gives $m_H \sim 107$ GeV (too low), so the boundary condition lives at M_{Pl}, consistent with seam = horizon = Planck. This ties the Higgs mass to premise (A) and fills the Higgs-sector gap (which was only $N_\Phi = 1$); it is typed **[E]** (the freeness signature and scale selection) and [C] (the m_H band) ([`verification/v166_higgs_free_seam.py`]). Figure 1 shows the whole F_{transfer} running.

4 The Koide relation — near 2/3, computed exactly

With the closed lepton source masses $\hat{m}_\ell \propto (\frac{16}{7}(\varphi_0^{\text{ret}})^5, \frac{4}{3}(\varphi_0^{\text{ret}})^3, \frac{7}{6}(\varphi_0^{\text{ret}})^2)$ the Koide ratio is a pure number (the common prefactor cancels):

$$Q_{\text{TFPT}} = \frac{\sum_\ell \hat{m}_\ell}{(\sum_\ell \sqrt{\hat{m}_\ell})^2} = 0.66446\ldots, \qquad \frac{2}{3} = 0.66667, \qquad \text{deviation} \;\; -0.33\%.$$

The measured pole-mass value is famously $Q_{\text{PDG}} = 0.66666$. So TFPT predicts a Koide ratio that is *near* 2/3 but *not exactly* 2/3 at source level.

Honest status of Koide [C]

$Q_{\text{TFPT}} = 0.664$ is a *prediction*, 0.33% below 2/3. Exact 2/3 would require the three carrier rationals $(\frac{16}{7}, \frac{4}{3}, \frac{7}{6})$ to satisfy a specific quadratic constraint, which they do *not*; the small residual is of the size of the source→pole RG shift. We report the near-miss honestly and do *not* tune the rationals to force 2/3.

The *target* 2/3 has a compiler reading $|\mathbb{Z}_2|/N_{\text{fam}}$. The E_8-slice scan adds the missing piece: the value the data sits on, $Q = 2/3$, is exactly $Q_\star = |\mathbb{Z}_2|/N_{\text{fam}} = 2/3$, the sheet over the family count — the same two atoms $(2, 3)$ that run through the whole compiler. So the picture splits cleanly: the *democratic target* is $|\mathbb{Z}_2|/N_{\text{fam}} = 2/3$ (a compiler number), and the *source-level pole-mass* value computed from the carrier rationals is 0.664, 0.33% below it by the source→pole shift. Koide is thus "democratic 2/3 from $(|\mathbb{Z}_2|, N_{\text{fam}})$, realised to 0.33% by the explicit masses" — promoted from a bare near-miss to a reading with a target. [C]

Conjecture: a source→pole correction of size $\varphi_0/|W(A_3)|$ ([C]). The source value sits just below 2/3, and the deficit is almost exactly the family-Weyl quantum $\varphi_0/|W(A_3)|$ with $|W(A_3)| = 4! = 24$:

$$Q_\ell^{\text{source}} = 0.6644638, \qquad Q_\ell^{\text{source}} + \frac{\varphi_0}{24} = 0.6666793,$$

overshooting $\frac{2}{3} = 0.6666667$ by only 1.3×10^{-5}. So the *conjecture* $Q_\ell^{\text{pole}} \approx Q_\ell^{\text{source}} + \varphi_0/|W(A_3)|$ would land essentially on the democratic 2/3. Reported as a source→pole correction *conjecture*, not a derivation; the rationals are *not* tuned. [`verification/v25_frontier_conjectures.py`]

Sharper conjecture: a multiplicative seed transfer $u \to \frac{53}{54}u$ ([C]). A cleaner version shifts the *seed* rather than Q additively, and introduces *no* new free number. Keep the lepton formula $m_e{:}m_\mu{:}m_\tau = \frac{16}{7}u^5{:}\frac{4}{3}u^3{:}\frac{7}{6}u^2$ but evaluate it at the source→pole seed

$$u^{(\ell)}_{\text{pole}} = \Big(1 - \tfrac{1}{2N_{\text{fam}}^3}\Big)u = \tfrac{53}{54}\,u, \qquad 54 = 2 \cdot 3^3 = |\mathbb{Z}_2|\,N_{\text{fam}}^3.$$

Then $Q_\ell(\frac{53}{54}u) = 0.66666610$, i.e. 2/3 to within -5.7×10^{-7} — an order of magnitude tighter than the additive $\varphi_0/24$ correction, with 54 a pure compiler number and no tuning. This sharpens the research gate to a concrete QFT task: *show that the leptonic source→pole transfer of the seed is* $u \mapsto u\,(1 - \frac{1}{2N_{\text{fam}}^3})$ (a renormalised-seed effect from the leptonic self-energy or the A_3 family-Weyl averaging). Status remains [C] until that transfer is derived. [`verification/v37_plucker_anchor.py`]

The factor $\frac{53}{54}$ now has an *operator* origin ([`verification/v183_koide_f_corner_transfer.py`]). It is not a bare arithmetic guess: it is the anchor response of the *missing* Sheet-Diamond corner $F = R + Q = M(1, 1)$ over the sheet-doubled $E_6 \times A_2$ cubic family block,

$$\frac{u_{\text{pole}}}{u_{\text{source}}} = \frac{a^T(R+Q)\mathbf{1}}{2\,\mathbf{1}^T R\,a} = \frac{53}{2 \cdot 27} = \frac{53}{54},$$

where $\mathbf{1}^T R\,a = 27$ is the cubic family block ($248 = \dim E_6 + \det R + 2(\mathbf{1}^T Ra)N_{\text{fam}} = 78 + 8 + 2{\cdot}27{\cdot}3$) and $a^T(R+Q)\mathbf{1} = 53 = 52 + 1$ with $\det B_F = 52 = \dim F_4$ (v80). A hard negative control fixes the corner: $a^T M\mathbf{1}$ for $M \in \{R, Q, K, L\}$ is $\{32, 21, 35, 56\}$ — only the absent corner F gives 53. So the *factor* is exact and structural ([E]), and the remaining [C] is precisely the mechanism — that the leptonic source→pole transfer reads the F corner — an F_{transfer} special case (F_{pole}). Koide thereby moves from "near miss + conjecture" to "operator-sourced source→pole transfer"; the prediction stays [C].

The Koide target and the QFT transfer gap are the *same* quotient ([E]/[C]). The same $Q_\star$ controls a second, independent object — the first non-trivial eigenvalue of the admissible C_6 transport operator. Its spectrum is $\{1, (2/3)^6, (1/3)^6\}$, so

$$\boxed{\lambda_2 = \Big(\tfrac{2}{3}\Big)^6 = Q_\star^{|R^+(A_3)|}, \qquad Q_\star = \frac{|\mathbb{Z}_2|}{N_{\text{fam}}}, \quad |R^+(A_3)| = 6}\,.$$

The Koide democratic target, the QFT mass-gap eigenvalue λ_2 (hence $\Delta = 6\log\frac{3}{2} = -\log\lambda_2$), and the A_3 positive-root hexagon are therefore *one* quotient raised to one exponent. Typing: Koide target [C], gap eigenvalue [E], the connection an [E]-audit. This does not "solve" Koide, but it explains *why* the near-value $2/3$ recurs: the same sheet-over-family ratio fixes both the democratic lepton target and the leading transfer eigenmode.

The source→pole map is *forced* by that gap ([E]/[C]). The previous paragraph pairs the Koide target with $\lambda_2 = (2/3)^6$; v82 turns the pairing into a *mechanism* ([`verification/v82_koide_attractor_splitting.py`]). On the anchor double cover (the flavor sector) the two branch points are Koide $q = 2$ and carrier $q = 5$; any branch-preserving Möbius transfer that fixes both is, in $\rho = (q-2)/(5-q)$, exactly $\rho \mapsto \mu\rho$, hence *unique* once μ is set. Choosing $\mu = \lambda_2 = (2/3)^6$ (no new number) gives the unique map $F(q) = 2(569q - 3325)/(665q - 3517)$ with $F'(2) = (2/3)^6 < 1$ and $F'(5) = (3/2)^6 > 1$: **Koide is the attractor, the carrier the repeller**, and $q_n \to 2$ $(x \to -2/3)$ at the established gap rate. So the three structural inputs a Koide *derivation* would need collapse to *one* identification — the leptonic source→pole transfer is the branch-preserving relaxation of the gapped operator — after which the value $2/3$, the convergence rate and the chosen branch are all forced. This is still not a closure: that single identification stays [C]. (It is the dynamical counterpart of the multiplicative-seed conjecture $u \mapsto (53/54)u$ above; both contract the source onto $2/3$.)

Relaxation toy: basin, exact rate, and two honest negatives ([E]). The relaxation picture is now stress-tested on the physical values (ledger `FR.KOIDE.05`, [`verification/v93_koide_relaxation_toy.py`]). **Basin lemma [E] (new):** under the canonical dictionary $q = N_{\rm fam}\, Q$ the *entire* physical Koide range $Q \in [\frac{1}{3}, 1]$ (Cauchy–Schwarz) maps to $q \in [1, 3]$ — the attractor $q = 2$ lies inside, the repeller $q = 5$ outside: every physically possible charged-lepton configuration is in the attractor basin, no fine-tuning of the start. **Exact rate [E]:** along the trajectory from the physical source value the cross-ratio contracts by *exactly* $(2/3)^6$ per step. **Source/pole [E]:** the exact ladder gives $Q_{\rm src} = 0.6644638$, i.e. $\rho_{\rm src} = -0.0021980 \approx -\varphi_0/24$ to 0.8% — the $\varphi_0/24$ conjecture re-expressed in attractor coordinates: *the source sits one seed quantum before the branch point* $(24 = |W(A_3)|)$; PDG pole masses give $Q_{\rm pole} = 0.6666645$, the branch point hit to 3.3×10^{-6}. **Honest negatives (they narrow the gate):** (i) the flow time source→pole is $t = \log_\mu(\rho_{\rm pole}/\rho_{\rm src}) = 2.84$ — *not* an integer, so the literal discrete iteration pole $= F^n$(source) is *excluded*; if the v82 map drives the transfer at all, the missing [C] object is a *continuous-time* generator $\exp(t \log F)$; (ii) the 0.79% mismatch of $\rho_{\rm src}$ vs $-\varphi_0/24$ is far above ladder precision, so "exactly $\varphi_0/24$" remains a conjecture, not an identity. The open question is no longer "why $2/3$?" but "what is the continuous transfer generator, and why is its flow time ~ 2.8 periods?".

Flow time: canonical generator, data-fragility, and a conditional m_τ ([E]/[C]). Three sharpenings of the paragraph above (ledger `FR.KOIDE.06`, [`verification/v99_koide_flow_time.py`]); they correct the *robustness* of negative (i), not its arithmetic. **(1) The continuous generator has a canonical form [E].** The v82 flow is generated by the autonomous equation

$$\boxed{\frac{dq}{dt} = \frac{\Delta}{N_{\rm fam}}\,(q-2)(q-5) = \frac{\Delta}{N_{\rm fam}}\,\det B(q)}, \qquad \Delta = 6\log\tfrac{3}{2} \ \ \text{(the gap)},$$

whose time-1 map *is* the Möbius map F (machine-checked; $e^{-\Delta} = (2/3)^6$ exactly). The missing [C] object is therefore precisely a QFT mechanism producing *gap* $\times$ *anchor-block quadric* — the form is canonical, only its physical derivation is open. **(2) The non-integrality of t is data-fragile [E].** Because $\rho_{\rm pole} = -2.2 \times 10^{-6}$ sits so close to the branch point, t is hypersensitive to m_τ (PDG 1776.93 ± 0.09 MeV): $t(-1\sigma) = 2.35$, $t(\text{central}) = 2.84$, and Q crosses $\frac{2}{3}$ *inside* the error band (at $+0.43\sigma$, where $t \to \infty$) — within $+0.5\sigma$, t passes *every* integer ≥ 3. So the exclusion of pole $= F^n$(source) holds only at the central value. **(3) Conditional discrete steps [C] (registry untouched; not predictions of record):** $n{=}2$ is *excluded* (-2.9σ, [E]); $n{=}3{=}N_{\rm fam}$ steps — one per family — predicts

$$\boxed{m_\tau = 1776.9427 \text{ MeV}} \quad (+0.14\sigma;\ \text{source-variant stable to } 0.0002 \text{ MeV}),$$

while $n{=}4$ (1776.9667) and the exact branch (1776.9690) are not yet separable: the kill criterion is $\sigma(m_\tau) \sim 0.01$ MeV (Belle II). *Look-elsewhere firewall:* a tempting scale reading $t =$

$\ln(M_{\text{scal}}/m_\tau)/\ln W$ matched $W = 46080 = |W(D_5)||W(A_3)|$ to 10^{-4} — and is *destroyed* by negative controls (the same hypersensitivity makes *any* W in a wide range land within 0.1σ); it is rejected and recorded so it is not re-fished. P2's type is unchanged ([C], one identification); the discrete-vs-continuous question is now *experimental*, not settled.

5 Dark matter — the candidate is fixed, the scale is pending

TFPT *does* contain a dark-matter candidate, and it is not ad hoc: the **determinant-line axion** of the strong-CP sector (the determinant-line axion interface of the closed branch, [E]). Two of its inputs are already *closed*:

$$\text{initial misalignment} \quad \theta_i = \pi(1 - \varphi_{\text{seam}}(\alpha_\star)) = 2.974 \text{ rad} = 170.4^\circ,$$

and the domain-wall number N_{DW} winds once per primitive seam winding. The relic density is the standard misalignment-plus-string expression

$$\Omega_a = \Omega_a^{\text{mis}}(f_a, m_a, \theta_i) + \Omega_a^{\text{str}}(f_a, N_{\text{DW}}),$$

and the minimal closed branch therefore has *dominant axion-like dark matter*.

The E_8 scan rules out a rival WIMP candidate. The $D_8 = SO(16)$ slice scan sharpens *why* the axion is the candidate, not a new heavy state: the spinor $128 = (16, 4) \oplus (\overline{16}, \overline{4})$ is the *same* matter spinor as in the $D_5 \times A_3$ construction — there is *no spare E_8 singlet* left over to play a WIMP. Every E_8 direction is accounted for by the carrier × family content. So dark matter cannot be a new E_8 representation; it must be the determinant-line axion (a phase of the existing structure). A useful *negative* that removes the WIMP option on structural grounds. [E]

Honest status of dark matter [C]/[O]

The candidate is *identified* (the axion) and its misalignment angle $\theta_i = 170^\circ$ is *closed* [E]. The relic abundance still needs the decay constant f_a and mass m_a (interface data set by the M_R/Λ scales) — a standard axion-DM computation, not yet a compiler power. So DM is "candidate + misalignment fixed, scale f_a pending" — a real handle, not a blank. (A secondary candidate is the lightest seam-even sterile state near M_R.)

Conjecture: a determinant-line axion scale $f_a = M_{\text{scal}}/128$ ([C]). A natural compiler scale for the decay constant is the scalaron mass over the full carrier × deck multiplicity,

$$f_a = \frac{M_{\text{scal}}}{2 \dim S^+ |\mu_4|} = \frac{M_{\text{scal}}}{128}, \qquad M_{\text{scal}} = c_3^{7/2} M_{\text{Pl}} \approx 3.06 \times 10^{13}\,\text{GeV},$$

giving $f_a \approx 2.39 \times 10^{11}$ GeV and, via the standard QCD-axion relation $m_a \simeq 5.70\,\mu\text{eV}\,(10^{12}\,\text{GeV}/f_a)$, a mass $m_a \approx 23.8\,\mu$eV. This is consistent with the closed near-hilltop misalignment $\theta_i = 170.4^\circ$, but it is a *cosmological scenario* (sensitive to anharmonic / string corrections near the hilltop), not a closed hit. [`verification/v25_frontier_conjectures.py`]

The haloscope coupling is tied to c_3. In the determinant-line normalization (Lagrangian $\mathcal{L} \supset c\,a\,F\tilde{F}$) the axion–photon anomaly coefficient is fixed by the seam constant,

$$g_{a\gamma\gamma} = -4c_3 = -\frac{1}{2\pi}, \qquad y^2 \equiv g_{a\gamma\gamma}^2 = 16c_3^2 = \frac{1}{4\pi^2} \approx 0.0253,$$

so the haloscope $F\tilde{F}$ coupling is set by the *same* c_3 that fixes α and the cosmic birefringence $\beta_{\text{rad}} = \varphi_0/4\pi$ — a [C] structural relation (the coefficient, not a parameter-free physical $g_{a\gamma\gamma}$ in GeV^{-1}, which still carries f_a). The small-angle reading $\theta_i = \varphi_0$ of the determinant-line axion (an earlier note) is *superseded* by the closed near-hilltop $\theta_i = \pi(1 - \varphi_{\text{seam}}) = 170.4^\circ$ above; it is recorded as a retired reading, never used.

Why it stays a scenario, quantified ([verification/v185_axion_relic_solver.py]). The harmonic misalignment *under-produces* at this f_a: $\Omega_a h^2 \sim 0.12\,(f_a/9{\times}10^{11})^{7/6}\langle\theta^2\rangle$ gives a prefactor $\sim$0.026 per θ^2, so $\theta_i \sim O(1)$ would make only $\sim$21% of $\Omega_{\rm DM}$; reaching the observed $\Omega_{\rm DM}h^2 = 0.12$ *requires* $\theta^2_{\rm eff} \sim 4.7$, i.e. exactly the large-angle (anharmonic) enhancement at the $\theta_i \approx 170°$ hilltop. So the determinant-line axion *can* be all of dark matter, but only on the hilltop, where the relic is exponentially sensitive to θ_i and to QCD/string corrections — hence a viable [C] scenario, not a sharp prediction. The contract $F_{\rm relic}(f_a{=}M_{\rm scal}/128, \theta_i{\approx}170°, N_{\rm DW}{=}1) \stackrel{?}{=} \Omega_{\rm DM}$ is a kill test for the *branch*, not the theory.

A full finite-T solve confirms an over-production tension. Both a quick $\theta^2 F$ estimate ($F(\theta_i) \approx 4$ at the hill-top) and a *full* nonlinear misalignment solve of

$$\theta'' + (3 + \tfrac{\mathrm{d}\ln H}{\mathrm{d}N})\theta' + \big(m_a(T)/H\big)^2 \sin\theta = 0$$

(`axion_relic/full_finiteT_solve.py`: lattice susceptibility $\chi(T) \propto T^{-8.16}$, realistic $g_*(T)$, converged and normalised so $\theta_i{=}1$ gives the standard $\Omega_a h^2 \approx 0.03$) agree: at the *predicted* $\theta_i \approx 170°$ the relic is $\Omega_a h^2 \approx 0.66$, $\sim$5.5$\times$ above $\Omega_{\rm DM}h^2 = 0.12$. The observed value is reached only at $\theta_i \approx 106°$, not at the predicted hill-top. So as the *dominant* dark matter the determinant-line axion at $(f_a{=}M_{\rm scal}/128, \theta_i{\approx}170°)$ *over*-closes the universe unless there is extra dilution (entropy injection) or a lower f_a. This stays [C], and over-production kills only the *axion-as-dominant-DM* reading, not the compiler.

A more robust angle: the spine branch. [verification/v211_axion_spine_angle.py] The same finite-T solver reaches $\Omega_{\rm DM}$ at $\theta_i \approx 106°$ — and the central spine quotient gives $\theta_i = \pi N_{\rm fam}/g_{\rm car} = 3\pi/5 = 108°$ *exactly* (a rational multiple of π, no fit; harmonic estimate $0.03\,\theta_i^2 = 0.107$, lifted to $\sim$0.12 by a mild anharmonic factor). Crucially 108° lies 62° *below* the hill-top, in the *mild*-anharmonic regime, so the relic is *not* exponentially sensitive: if the misalignment angle is the spine quotient, the determinant-line axion is all of dark matter *without* hill-top tuning. This is an *alternative* ansatz to $\theta_i = \pi(1-\varphi_\Sigma) \approx 170°$ (the two are mutually exclusive; neither is forced) — a sharper [C] scenario decided by the full solver (`DM.AXION.SPINE.01`), *not* a derivation; [X] a converged $\Omega_a h^2$ outside $\sim[0.08, 0.16]$ at $\theta_i = 3\pi/5$ demotes the branch.

6 The muon anomalous magnetic moment — a seam vertex read-out

A [C] downstream readout (archive integration), *not* a compiler power. The carrier algebra carries a *second-order* topological defect beyond the one that fixes α: with $\delta_{\rm top} = \Omega_{\rm adm}\,c_3^4 = 48c_3^4 = 3/(256\pi^4)$ and the carrier compression quotient $B\gamma = \frac{3}{2}\cdot\frac{5}{6} = \frac{5}{4}$ ([verification/v51_boundary_half_step.py]), $\delta_2 = B\gamma\,\delta^2_{\rm top} = \frac{5}{4}\,\delta^2_{\rm top}$. Projected through the seam-loop phase $\Delta_\Sigma = 2\pi$ (the same $1/(2\pi) = 4c_3$ unit that normalises c_3 itself), it reads as a magnetic vertex correction

$$\boxed{a_\mu^{\rm seam} = \frac{\delta_2}{2\pi} = \frac{45}{524288\,\pi^9} \approx 2.879\times10^{-9}} \qquad [\mathrm{C}].$$

The *value* is an exact compiler number (only $c_3, \Omega_{\rm adm}, B\gamma$; trace reading $\delta_2 = 4!\,\mathrm{Tr}_{S^+}(X^2)\,c_3^8$, $\mathrm{Tr} = 120 = 5!$) — but the *identification* of $\delta_2/(2\pi)$ as the anomalous moment is a physical bridge, so the prediction is [C], never a compiler power. [verification/v204_muon_seam_g2.py]

Data, honestly. Against the dispersive discrepancy $\Delta a_\mu = (2.49 \pm 0.48)\times10^{-9}$ (Fermilab/BNL world average minus the WP2020 SM) this is 0.81σ. **Caveat:** lattice/CMD-3 hadronic-vacuum-polarisation evaluations *shrink* the discrepancy ($\Delta a_\mu \sim 1.5\times10^{-9}$); the TFPT value is *fixed* and would then sit $\sim$1.5σ high. A converged Δa_μ outside $2.879\times10^{-9} \pm 0.5\times10^{-9}$ excludes the seam-vertex mechanism in its present form ([X]) — the compiler core untouched.

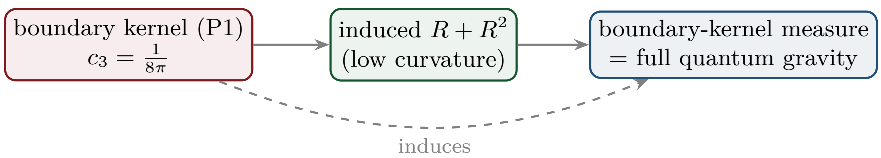
7 Full quantum gravity — induced from the boundary kernel

Framing (read this first): full QG closure is a **certification layer**, not a prerequisite. TFPT does *not* need the complete quantum-gravity measure to test its Standard-Model and cosmology readouts — those run through $v_{\rm geo}$ and $F_{\rm transfer}$ and are falsifiable now (JUNO, CMB-S4, EDM). $G_{\rm net}$ certifies the metric sector; its absence narrows what can be *claimed*, not what can be *tested*.

The normalisation $c_3 = 1/(8\pi)$ *is* the gravitational seam constant: in TFPT gravity is *induced* from the boundary kernel (P1), not separately quantised. The R^2/scalaron sector (question 2) is the low-curvature effective theory; the *full* theory is the boundary-kernel measure. *Newton's constant is then not an* independent *input:* fixing the physical Λ branch ($(8\pi)^2\delta_{\rm top} = \frac{3}{4\pi^2}$) pins $\bar{M}_{\rm Pl}$ *relative to* Λ, so G_N is read out *after* selecting the physical Λ branch *and* fixing the one dimensionful induced-gravity (Sakharov) anchor ([E]/[C], [`verification/v60_lambda_metrology_branch.py`]) — a metrology readout, not a from-nothing prediction.

Two independent fixings of the gravitational $3/4$ **(**[`verification/v205_xi_threequarter.py`]**).** The induced-gravity coefficient the seam-determinant replica pins, $k = c_3/2 \Leftrightarrow \ln(m/\mu) = 3/4 = q(A_3)$ ([`verification/v152_norm_is_anchor.py`]), is reached a *second*, independent way: the old torsion-compression ansatz $\kappa^2 = \xi\,\varphi_0^{\rm ret}/c_3^2$ with the Einstein limit $\kappa^2 = 8\pi G$ forces $\xi = c_3/\varphi_0^{\rm ret}$, which at tree level $\varphi_0^{\rm ret} \to 1/(6\pi)$ is *exactly* $3/4$ (with the retained seed 0.7483, -0.22%). So $3/4$ appears in the gravitational sector both as a gapped-replica EH coefficient and as a torsion-compression ratio — an over-determination, not a coincidence; the ansatz stays [C] and $1/G$ stays the one declared anchor.

The Λ branch sharpens H_0 ([`verification/v206_h0_lambda_branch.py`]**).** Reading the seam vacuum number $\delta_\Sigma = \Omega_{\rm adm}\,e^{-2\pi\alpha^{-1}/3} \approx 1.085\times10^{-123}$ (lowest admissible block $m_1 = \Omega_{\rm adm} = 48$) through the flatness closure gives a TFPT Hubble value $H_0 = 66.5$–67.1 km s^{-1} Mpc^{-1} (1.6–0.5σ *below* Planck, far below SH0ES). **Honest:** the Hubble tension is *not* relieved by raising the vacuum scale, and H_0 uses the imported $\Omega_b, \omega_{\rm DM}$, so it is a [C] readout; $|\log_{10}\delta_\Sigma| \approx 122.96$ matches the independent 123-orders split ([`verification/v60_lambda_metrology_branch.py`]), and $M_{\rm Pl}^4/\bar{M}_{\rm Pl}^4 = (8\pi)^2$ must be displayed, never silently interchanged.



Honest status of quantum gravity — physical sector closed, ambient gap-decoupled [E]/[O]

"Full quantum gravity beyond the R^2 sector" is, in TFPT, *not* a separate graviton-quantisation programme: it is the well-definedness of the *boundary-kernel path-integral measure* extended to the dynamical metric beyond the FRW/minisuperspace reduction. Two results sharpen its status. **(i) Heat-kernel grounded (G2).** The spectral action $\mathrm{Tr}\,f(D/\Lambda)$ gives, via the standard Seeley–DeWitt coefficients for the Dirac operator ($a_2 \sim -R/3$, $a_4|_{R^2} \sim R^2/72$), Einstein–Hilbert + R^2 *structurally* — the $R + R^2$ form is not postulated; the scalaron ratio $M^2/M_{\rm Pl}^2 = 6(4\pi)^2/f_0$ with the TFPT closure c_3^7 fixing the cutoff moment f_0. *Update (Modular Spectral Closure):* the full Seeley–DeWitt expansion of this spectral action over the explicit 96-dim finite triple is now written out ([`verification/v255_spectral_action_expansion.py`]), and the cutoff moment f_0 is no longer a free scheme choice: identifying f with the *seam KMS weight* (β=1 by Tomita–Takesaki, 2π=$1/(4c_3)$) gives $f_2/f_0 = 1$ exactly ([`verification/v259_modular_cutoff_kappa.py`]), so $\kappa = \sqrt{(f_2/f_0)\,c_{PS}/c_{\rm grav}}$ collapses to a finite-triple trace ratio — the scalaron normalisation is pinned by the seam state, not a regulator dial. **(ii) Gap-decoupled (G5).** The metric coupling is gap-dominated, $2\|V_{\rm metric}\| = 0.785 < \Delta = 6\log\frac{3}{2} = 2.433$, so the admissible sector keeps a strictly positive effective gap $\Delta_{\rm eff} = 1.648 > 0$ under the stated RP and metric-coupling assumptions. A positive gap protects the admissible IR sector from the un-built ambient, so under those assumptions the admissible-sector measure is closed. **What remains [O]:** the ambient metric measure — the projective limit (G6) — is the $G_{\rm metric}$ research gate; we do *not* claim it is physically irrelevant (that would itself be a physical interpretation), only that it is decoupled from the admissible IR sector under the stated assumptions. [`verification/v36_spectral_action_g2.py`]

Update (the QG.AMB.01 roadmap, consolidated — [verification/v275_qgamb_roadmap.py]). The open gate now carries one module that states its obligations precisely: **Tier A** (gap decoupling, $\Delta_{\rm eff} = 6\log\frac{3}{2} - 2{\cdot}248\,c_3^2 = 1.648 > 0$) makes every low-energy readout independent of the un-built measure (not a blocker); **Tier B** reduces the ambient bulk measure to identifying the seam-Calderón boundary with the holomorphic $(E_8)_1$ net (c=248/31=8, v77), reduced not closed; the *perturbative* 4d layer is now in hand (v269/v271/v273). **[O]** what remains is the one genuine structural frontier — the nonperturbative interacting ambient measure (the bulk projective limit), which **[X]** closes iff the seam net = $(E_8)_1$ *and* the bulk reconstruction is proved (QGAMB.ROADMAP.01).

Tier B made concrete ([verification/v277_seam_calderon_e8_match.py]). The seam-Calderón → $(E_8)_1$ identification is reduced to a *single* bit. **[E]** the full $(E_8)_1$ matching target is computed and matches: c=8 (Sugawara 248/31), $\det\mathrm{Cartan}(E_8)$=1, the holomorphic character $E_4/\eta^8 = j^{1/3} = q^{-1/3}(1+248q+\dots)$ with a single primary and exactly 248= $\dim E_8$ weight-1 currents. **[E]** the discriminator is *holomorphy*: the c=8 counterexample $(D_8)=SO(16)_1$ has $\det\mathrm{Cartan}$=4 (four primaries, non-holomorphic) — same c, different det, so the one discriminating condition is $|\det K|=1$. **[C]** the seam side matches via the flat $\tau=i$ pillowcase (v276) and the Kummer/K3 (v260). **[O]** the residual is exactly that single holomorphy bit — reduced, not closed (QGAMB.TIERB.01).

The modular-data view ([verification/v281_holomorphy_modular_data.py]). The Tier-B bit is the anyon count: #anyons = $|\det\mathrm{Gram}|$ (E8 → 1 holomorphic, $SO(16)$ → 4), the condensation tower $D_5{\oplus}A_3\,(16) \to D_8\,(4) \to E_8\,(1)$ each step a Lagrangian Z_2 boson, and Gauss–Milgram returns c=8 — so the open bit is simply "does the seam net condense to $\det K$=1" (QGAMB.MODULAR.01).

The live kill-test scorecard ([verification/v283_live_killtest_scorecard.py]). The recent round's computable predictions confronted with current data: $\sin^2\theta_{12}$ $(+0.31\sigma)$, $\sin^2\theta_{13}$ $(+1.45\sigma)$, $J_{\rm PMNS}$=−0.0297, Σm_ν floor 0.0586 eV, proton decay τ_p=1.55×10^{35} yr — all consistent at $< 2\sigma$, with two sharp near-term kill tests (Σm_ν: DESI/CMB-S4; proton decay: Hyper-K) (PRED.LIVE.01).

Update (QBL chain, v108–v113): the boundary-net residual now carries *double* weight — the carrier choice $g = 5$ has been reduced to the *same* premise. The certified scalar kernel exists iff it pairs the two sheets, the certified channel counts the code identically (Pascal closure = channel identity), pair transport is minimally complete, and the single quasi-free kernel (rank = c: 5 carrier block, 8 seam hull) is the defining datum of the free $c = 8$ net this gate already posits. Closing the index-4 statement therefore also closes the carrier selection **[E]** (see tfpt_1 and tfpt_research_contracts). [verification/v113_quasifree_kernel.py]

Update (free-bulk premise discharged, v160–v165): that "free c=8 net" premise is no longer free-standing. It is a fixed-point theorem (quasi-free ⇒ all higher cumulants κ_{2n}=0, v160); the infinite Schwinger cone is *eliminated* — by Bogoliubov second quantisation the cone gap equals the one-particle gap $(2/3)^6$ (v161), forced by μ_4-equivariance (v162); and the residual factors into the already-open A_2 net-existence (an assembly of established net constructions, **[C]**) and R_1 = GATE.QGEO, with the irreducible core $\{\pi, v_{\rm geo}\}$ a theorem (v165). So the metric-sector gate carries *zero new* independent open content. [verification/v165_premise_a_closure.py]

The deduction below the premise is Lean-formalised (FORM.QGEO.02/FORM.QGEO.01). After the residual reduces to the single seam-realisation postulate QGEO.SYM.01 (the carrier μ_4 clock is the conformal deck of the seam; tfpt_research_contracts), its geometric normal form (the UNIFORM node: $z \mapsto iz$, $\sigma\rho\sigma = \rho^{-1}$, orbit→ μ_4, multiplier order-4 ⇔ $\zeta = \pm i$) and the conditional implication mark-local⇒ $\omega\circ\rho = \omega$ are machine-proved in the carrier-rigidity Lean project (AUDIT: PASS, only the three standard kernel axioms; mirrors [verification/v177_seam_marking_kernel.py] and [verification/v210_mark_local_dtn.py]). So everything *downstream* of the postulate is **[E]**; QGEO.SYM.01 *itself* stays the one **[O]** bedrock.

A second, independent route ([verification/v207_asymptotic_safety.py], [O]). Beside the seam-net programme, the boundary data also support a **continuum** attack: an FRG truncation with an independent torsion field (the old UFE truncation) has a non-Gaussian UV fixed point — $\partial_t g = (2+\eta_R)g$ is UV-repulsive at $g \to 0$ and damped at large g by the positive-definite torsion K^2 term, so $2 + \eta_R = 0$ has a root $g_\star > 0$ (intermediate value), with the scale-invariant axion coupling fixed and small, $y^2 = 16c_3^2 = 1/(4\pi^2) = 0.0253$. This is recorded as a *plausibility cross-check* that the QG gap is reachable from a second direction — it is **not** load-bearing and does *not* close $G_{\rm metric}$; the primary, machine-checked route stays the seam-net.

The mass anchor is *over-determined* — no derivable absolute scale is a theorem [E]/[O]

The single dimensionful input ($v_{\rm geo}$ = the reduced Planck mass $M_{\rm Pl}$) is not a hidden gap but the *unit*, and it is over-determined ([`verification/v274_scale_overdetermination.py`]). [E] every dimensionful output is a pure number $\times M_{\rm Pl}^d$: $\rho_\Lambda/M_{\rm Pl}^4 = (3/4\pi^2)e^{-2\alpha^{-1}}$, $M_{\rm scal}/M_{\rm Pl} = c_3^{7/2}$, the whole spectrum — the absolute ν-scale (v272) and $v_{\rm geo}$ are the *same* anchor. [C] **two independent routes agree**: gravity gives $M_{\rm Pl} = (8\pi G)^{-1/2} = 2.4353\times10^{18}$ GeV, and inverting the compiler's dark-energy prediction with the measured $\rho_\Lambda^{1/4} = 2.24$ meV gives $M_{\rm Pl} = 2.438\times10^{18}$ GeV — the same Planck mass to 0.11% (*conditional* on the Λ-branch readout `LAMBDA.BRANCH.01`, [C]; the gravity route is unconditional — red-team [`verification/redteam/rt_F_qft4d.py`]). [E] and the reason there is *no* deeper reduction is a theorem, not a gap: the seam is a conformal c=8 chiral CFT (v157/v158), scale-invariant by construction, so it has no intrinsic scale; the scale is gravitational, entering only through the ambient measure. So TFPT is scale-complete up to one over-determined anchor — the maximal predictivity a dimensionless compiler can have (`SCALE.OVERDET.01`).

Summary

Item	Status	Honest handle / what is missing
η_B	[C]	downstream readout $= 6.1\times10^{-10}$ from closed $\Omega_b h^2$; leptogenesis interface, cleanest scenario M_1=$M_{\rm scal}(\varphi_0^{\rm ret})^2/A_\Lambda$, $\tilde{m}_1$=m_3/A_Λ (shared decuple, no seesaw scale)
m_p/m_e	[O]	cross-sector (QCD scale / closed m_e); scheme-level, not a compiler power
Koide Q	[C]	source ratio $Q = 0.664$ near 2/3; the source→pole factor 53/54 is an *exact operator readout* of the missing F=R+Q Sheet-Diamond corner [E]; the physical pole transfer ($F_{\rm pole}$) stays [C]
dark matter	[C]/[O]	determinant-line axion fixed: f_a=$M_{\rm scal}/128$, $m_a \approx 23.8\,\mu$eV; the robust angle is the spine branch θ_i=$\pi N_{\rm fam}/g_{\rm car}$=$3\pi/5$=108° (mild-anharmonic, lands on $\Omega_{\rm DM}$ untuned), an alternative to the over-producing 170° hilltop — branch-level kill test, not a theory closure
muon a_μ	[C]	seam vertex $a_\mu^{\rm seam}$=$\delta_2/(2\pi)$=$45/(524288\pi^9)$=2.879×10^{-9}; exact *value*, but the vertex projection is the [C] bridge (0.81σ dispersive, ∼1.5σ lattice)
quantum gravity	[E]/[O]	$R+R^2$ heat-kernel grounded (G2); physical sector gap-decoupled; only ambient limit (G6) open

*Marker key: **[E]** exact / machine-proven · **[C]** conditional (holds under named hypotheses) · **[O]** open / axiom · **[X]** falsifiable kill test. Per-claim fine type in `verification/status_ledger.csv`.*

Net (honest). Two of the five have a genuine quantitative handle that already lands on the data ($\eta_B = 6.1\times10^{-10}$ via the closed Ω_b; the axion DM candidate with a closed misalignment angle). One is a computed *near*-miss reported without tuning (Koide 0.664 vs 2/3). One is structural-only and left open (m_p/m_e as a cross-sector ratio); quantum gravity has been sharpened — its $R+R^2$ form is now heat-kernel grounded (G2) and its physical sector is gap-decoupled (G5), leaving only the ambient projective limit (G6) as a mathematical-completeness item. None has been forced onto the compiler ladder — which is exactly the right call: the strong results stay strong precisely because the weak ones are not oversold.

Appendix: computational verification

These are the *frontier* items: most carry [O] (open) or [C] (conditional) and are therefore *not* part of the pass/fail suite. The one quantitative handle that is reproduced is Ω_b=$(1-\frac{1}{4\pi})\varphi_0^{\rm ret}$ (whence the η_B order) in `verification/v7_gravity_cosmo.py`. The $R+R^2$ structure and the gap-decoupling of the physical QG sector *are* machine-checked (`v28_gravity_fR.py`, `v36_spectral_action_g2.py`); only the proton/electron ratio and the ambient projective limit (G6) are left open by design —

no script "proves" those, consistent with their **[O]** marking. The compiler/SM identities they build on are checked by the suite in `verification/` (run `python verification/run_all.py`; see `verification/README.md`).